

JAYANI SAMARAKOON

Undergraduate in Software Engineering

+94 716246625 | Kegalla, Sri Lanka | kiruusamarakoon@gmail.com | [github](#) | [dagshub](#) | [linkedin](#) | [portfolio](#)

PROFESSIONAL SUMMARY

Software Engineering undergraduate with hands-on experience in software development, machine learning, natural language processing (NLP), UI/UX design, and user interface development through academic and research projects. Proficient in Python, Java, SQL, data preprocessing, feature engineering, exploratory data analysis (EDA), predictive modeling, and web application development. Passionate about designing user-centered applications and building intelligent AI-powered solutions to solve real-world problems. Seeking an opportunity to apply technical, analytical, and creative skills in a collaborative and innovative environment.

EDUCATIONAL QUALIFICATIONS

BSc (Hons) in Software engineering

Sri Lanka Technology Campus (SLTC)

2023-Present

G.C.E. Advanced Level

K/MW Pinnwala central college

Year of 2022

Engineering technology stream: Engineering technology , Science for technology , Information Technology

PROJECTS

DeepShield: Explainable Multimodal Framework for Trust Assessment and Evidence-Based Fake News Detection (

Ongoing - Expected Completion: 2027)

- Developing an AI-based multimodal fake news detection framework using text, image, metadata analysis, and Retrieval-Augmented Generation (RAG).
- Integrating Explainable AI (XAI) techniques to provide transparent predictions and a Trust Score.
- Building a web-based prototype using FastAPI and React.js.
- Technologies: Python, PyTorch, Transformers (Hugging Face) OpenCV, CLIP, BERT, FAISS, FastAPI, React.js.

Medical Store Management System (Group)

June 2025 – Aug 2025

Responsibilities :

- Designed and developed responsive user interfaces using React.js.
- Created reusable React components for the web application.
- Designed the UI/UX to improve usability and user experience.
- Integrated frontend pages with the backend APIs.
- Collaborated with team members using Agile development practices.
- **Technologies:** React.js/CSS/HTML/JavaScript

End-to-End Customer Churn Prediction with MLOps (Individual)

Feb 2026 – Mar 2026

Built a production-ready MLOps pipeline that predicts customer churn for Telco customers.

- Cleaned and prepared 7,043 customer records by converting data types, handling missing values, encoding categorical variables, scaling features, and creating an 80:20 train-test split.

- Trained and compared 3 machine learning models (Logistic Regression, Random Forest, and XGBoost), selecting Logistic Regression as the production model with 81.97% accuracy and 0.8621 ROC-AUC.
- **Technologies:** Python, Scikit-learn, MLflow, DVC, Apache Airflow, Docker, FastAPI, DAGsHub

SMS SPAM Detection system (Group)

Juny 2026 – July 2026

Responsibilities:

- Developed and trained a Random Forest classifier for SMS spam detection.
- Built and optimized an LSTM deep learning model for sequential text classification.
- Performed feature engineering and hyperparameter tuning to improve model performance.
- Designed and developed a responsive web-based user interface for real-time SMS spam prediction.
- Compared machine learning and deep learning model performance using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- **Technologies:** Python, Random Forest, LSTM, Flask/FastAPI, HTML, CSS, JavaScript, Scikit-learn, TensorFlow

Hospital Appointment No-Show Prediction Using Machine Learning (Group)

August 2026

Responsibilities:

- Developed a machine learning solution to predict patient appointment no-shows using healthcare data.
- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA).
- Built and evaluated a **Logistic Regression** classification model.
- Conducted **Feature Importance Analysis** to improve model interpretability.
- Developed a **Streamlit web application** for real-time appointment no-show prediction.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, Matplotlib, Jupyter Notebook, Streamlit, Joblib

Mobile Application Development Project – LankaSmartMart (Group)

Jan 2026 – Feb 2026

- Designed the complete UI/UX prototype using Figma, including user-friendly screens and application workflow.
- Developed the Android mobile application using Android Studio by implementing the designed interfaces and core functionalities.
- Integrated the application with the **database** to enable real-time data storage and retrieval.
- Collaborated with team members using Git, GitHub, managing source code through version control and contributing code via commits and pushes.
- Worked closely with the development team throughout the Software Development Life Cycle (SDLC), including planning, UI/UX design, development, testing, and deployment.
- Participated in team discussions, problem-solving, and feature implementation to successfully complete the project.
- **Technologies:** Figma, Android Studio (Java/Kotlin), Firebase/MySQL, Git, GitHub

ACHIEVEMENTS & CERTIFICATES

- Certificate of Appreciation, Emerging Scholars' Research Programme - Certificate Phase - Faculty of Postgraduate Studies & Research, SLTC | 2025
- SQL (Basic, Intermediate) Certification — HackerRank | 2025

SKILLS

Technical Skills

- **Programming Languages** – Python, Java
- **Data Analysis & Visualization** – Pandas, NumPy, Excel, Power BI initiative to achieve project goals.
- **Big Data & Data Engineering** – PySpark, Apache ,Airflow, MLflow
- **Databases** – MySQL
- **Frameworks** – Flask, Flutter
- **Tools & Platforms** – Google Colab, Jupyter Notebook, Visual Studio Code, Docker, Jira
- **Version Control** – Git, GitHub, DAGsHub

Soft Skills

- **Team Collaboration & Leadership**
Works effectively in teams and takes the initiative to achieve project goals.
- **Project Management**
Plans, organizes, and completes tasks efficiently to meet deadlines.
- **Learning Agility**
Quickly learns new tools, technologies, and techniques to improve performance.

REFERENCES

Available Upon Request